

## **ENTI 333 L01**

### **Generative AI and Prompting**

#### **Course Outline Part A – Spring 2026**

|                         |                                                                                                  |
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| <b>Instructor</b>       | Dr. Mohammad Keyhani                                                                             |
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| <b>Office hours</b>     | By appointment only ( <a href="http://www.calendly.com/keyhani/">www.calendly.com/keyhani/</a> ) |
| <b>Website</b>          | <a href="http://d2l.ucalgary.ca">http://d2l.ucalgary.ca</a>                                      |
| <b>Lecture location</b> | SH 210                                                                                           |
| <b>Lecture times</b>    | 8:30am-5:30pm everyday May 11-15, 2026.                                                          |

Approximate schedule for each session:  
8:30-9:00am is warmup: open networking, co-working and reflection time. Lecture typically starts at 9:00am.  
10:20-10:30am refresher break.  
12pm-1pm is lunch break.  
2:20-2:30pm refresher break.  
4:30pm-5:30pm is cooldown: optional open networking, co-working and reflection time.

Please be in class no later than the end of break times.

**Prerequisites/Corequisites:** None

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| <b>Course Description</b> | Learn to use generative AI effectively by understanding the technology, and using prompting techniques, templates and tools. Emphasis on innovative and entrepreneurial applications through the use of generative AI tools and models directly, as well as indirectly through APIs. Involves use of scripting and no-code tools to build solutions with generative AI. Note that this course is merged with a graduate course (ENTI 633) but this document is for the undergraduate version of the course only. |
| <b>Course Modality</b>    | This is an in-person course, but some components such as lab exercises and group project may not require strict in-class presence at all times. There are no fully online or                                                                                                                                                                                                                                                                                                                                     |

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asynchronous lecture components, but the instructor may provide audio recordings of lectures after they are delivered.

**Course Objectives**

By the end of this course, students will:

1. Understand how generative AI works and identify relevant tools, trends, and implications for society
2. Demonstrate how knowledge of generative AI technology can be applied in the context of AI prompting.
3. Understand, analyze, and discuss the ethical issues around generative AI
4. Apply generative AI tools directly to design solutions and create value
5. Apply generative AI tools indirectly through APIs and scripting to design solutions and create value

Identify the problems that are most suited to tackling with generative AI as well as cases where it may be prudent to not use generative AI.

**Textbook and/or Other Materials and Resources**

**Students are required to have a laptop or similar computing device with them in all class sessions. Required readings** (if any) will be determined by the instructor and posted on D2L. Other recommended readings will be posted on D2L as well. There is **no required textbook** for this course, but the following are **recommended resources** (parts of which may be listed in required readings):

Kim, G., & Yegge, S. (2025). Vibe Coding: Building Production-Grade Software With GenAI, Chat, Agents, and Beyond. IT Revolution. <https://www.amazon.ca/Vibe-Coding-Building-Production-Grade-Software/dp/1966280025/>

Phoenix, J., & Taylor, M. (2024). Prompt engineering for generative AI: Future-proof inputs for reliable AI outputs. O'Reilly Media. <https://www.amazon.ca/Prompt-Engineering-Generative-AI-Future-Proof/dp/109815343X/>

Goss, K. (2023). Automate It with Zapier and Generative AI: Harness the power of no-code workflow automation and AI with Zapier to increase business productivity. Packt Publishing. <https://www.amazon.ca/Automate-Zapier-Generative-automation-productivity/dp/1803239840/>

The following online resources are recommended for learning about Generative AI:

- Microsoft course on “Generative AI for Beginners”  
<https://microsoft.github.io/generative-ai-for-beginners/#/>
- Microsoft course on “AI Agents for Beginners”  
<https://github.com/microsoft/ai-agents-for-beginners>
- DeepLearning.ai courses  
<https://www.deeplearning.ai/courses/>
- OpenAI Academy  
<https://academy.openai.com/>
- Anthropic courses  
<https://github.com/anthropics/courses>

**Statement on the use of generative AI:** The use of generative AI tools is required. Subscriptions to the **paid plans** of OpenAI ChatGPT or Anthropic Claude are recommended but not required. A paid plan on an AI coding tool like Replit.com is

highly recommended although it may be possible to survive the AI coding work in this course with only free plans on alternative AI coding tools.

Students are not required to cite AI tools (unless it makes sense to do so), but are asked to follow the “Read and Revise Until You Own (RRUYO)” rule because they will be held responsible for the content as explained here: <https://www.digitvibe.com/essence-of-writing/>

**Importantly, I do not expect to see deliverables that look purely AI generated, even when I ask you to use generative AI for the assignment. The point is to use generative AI to improve your work as much as possible, not to copy and paste blindly from it. Signs of blind copy pasting are penalized in grading.**

**Course  
Workload**

A block week course is an intense period of study. Students should expect to spend at least two hours per day outside of class time to work on this course.

**Grade Scale** The Haskayne School of Business endeavours to ensure consistency of final grades across courses and sections. Variations in distribution will always be considered by the instructor where called for by the performance in each individual class. The student does not have any 'right' to a certain grade but is responsible for earning grades. The instructor has unfettered discretion to evaluate student performance and assign all grades.

**Note to instructor:**

Please insert in table the numerical breakdown for your course. Use equal sign or >.

|           |   |      |                                                                                    |
|-----------|---|------|------------------------------------------------------------------------------------|
| <b>A+</b> | ≥ | 95.0 | Outstanding                                                                        |
| <b>A</b>  | ≥ | 90.0 | Excellent                                                                          |
| <b>A-</b> | ≥ | 85.0 | Approaching excellent                                                              |
| <b>B+</b> | ≥ | 80.0 | Exceeding good performance                                                         |
| <b>B</b>  | ≥ | 75.0 | Good performance                                                                   |
| <b>B-</b> | ≥ | 70.0 | Approaching good performance                                                       |
| <b>C+</b> | ≥ | 65.0 | Exceeding satisfactory performance                                                 |
| <b>C</b>  | ≥ | 60.0 | Satisfactory performance                                                           |
| <b>C-</b> | ≥ | 55.0 | Approaching satisfactory performance                                               |
| <b>D+</b> | ≥ | 52.0 | Marginal pass. Insufficient preparation for subsequent courses in the same subject |
| <b>D</b>  | ≥ | 50.0 | Minimal pass. Insufficient preparation for subsequent courses in the same subject  |
| <b>F</b>  | ≥ | 0%   | Failure. Did not meet course requirements.                                         |

**Grade Distribution**

| Due Date     | Assessment                                              | Weighting   | Course Outcomes Assessed |
|--------------|---------------------------------------------------------|-------------|--------------------------|
| May 17, 2026 | Individual: Class Participation & online forum activity | 20%         | 1,2,3,4,5                |
| May 11, 2026 | Individual: Lab 1                                       | 20%         | 1,2,4                    |
| May 13, 2026 | Individual: Lab 2                                       | 20%         | 1,2,3                    |
| May 17, 2026 | Group Project                                           | 40%         | 1,2,4,5                  |
|              | <b>Total</b>                                            | <b>100%</b> |                          |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>Missed Assessment Policy</b>   | <p>Students must follow the guidelines outlined in <a href="#">Part B</a> of the outline to request a deferral for missed work during the term, including quizzes, assignments, and exams.</p> <p>Students must coordinate with the instructor to seek a deferral for missed work during the term, including quizzes, assignments, and exams. Typically, deferrals are only granted in cases of illness, domestic affliction, or religious conviction, and are entirely at the discretion of the instructor. If a deferral is granted, an instructor may require a make-up assessment to be completed or transfer the weight of the missed assessment to another assessment in the course.</p>                                                                                                                                                                                                                 |
| <b>Late Policy</b>                | <p>Carefully read the course timetable and emails from the instructor for assignment due dates. <b>Late assignments will result in a penalty of 20% per day late</b>, including weekends and holidays, unless a valid exception has been made by the Instructor prior to the due date. Exceptions will only be granted, at the instructors' discretion, who will typically require presentation of suitable documentation. Assignments less than one day late are subject to smaller penalties at the discretion of the instructor. Note that an assignment submitted five or more days after the due date will be assigned a grade of 0%.</p>                                                                                                                                                                                                                                                                 |
| <b>Class Participation</b>        | <p>The class participation grade will be made up of an in-class and an online component. The in-class component involves completion of in-class exercises, professionalism in working &amp; communicating with your classmates and the instructor, as well as contribution to classroom discussions. The online component involves activity on the class discussion board(s) on D2L, including posting, commenting, and voting. See Appendix for details. Given that the online component is more easily trackable, it will be used as the main basis for participation grades, with the in-class component used as "additional consideration" for grading participation.</p> <p><b>Students are strongly advised to have a comprehensive and up-to-date LinkedIn profile. Note:</b> for this class you may need to sign up for many different web applications.</p>                                           |
| <b>Individual Lab Assignments</b> | <p>Lab assignment instructions are provided on D2L by the instructor and made available on the day of the assignment as indicated in the course timetable. Students are required to follow the instructions and submit the lab deliverables on D2L by the deadline indicated in the course timetable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Final Group Project</b>        | <p>See Appendix for details.</p> <p>Although some class time is allocated to group projects, please consider that your team will likely need to have meetings outside of class time. Students are expected to participate in class activities and the group project through communication with all team members, attendance at team meetings and execution of the assignment in a collegial manner. All group members should be present at group presentations, but not all members are required to present.</p> <p>The details of the group project can be found in the appendix to this course outline. All deliverables must be submitted as per the assignment instructions through D2L, and NOT through email.</p> <p>In the case of conflicts within your group, you are encouraged to try to resolve the conflict internally, although you may bring it up with the instructor as a last resort. If</p> |

the instructor finds enough evidence of social loafing or other indications that a group member has fallen short of their group work responsibilities, the instructor has full discretion to reduce the group project grade for that individual as they see fit or require remedial work by the individual.

**Final Exam**

There is no final exam for this course.

**Assessment of Writing**

Writing skills are not exclusive to English courses and, in fact, should cross all disciplines. The University supports the belief that throughout their University careers, students should be taught how to write well so that when they graduate their writing abilities will be far above the minimal standards required at entrance. Consistent with this belief, students are expected to do a substantial amount of writing in their University courses and, where appropriate, members of faculty can and should use writing and the grading thereof as a factor in the evaluation of student work. The services provided by the Writing Support, part of the Student Success Centre, can be accessed by all undergraduate and graduate students who feel they require further assistance.

**In this course, it is expected that you will always improve your writing using ChatGPT or other AI tools. It is also expected that you will NOT list AI as a co-author or acknowledge use of AI tools, because that could be seen as an offloading of responsibility. You are the only entity held accountable for your writing, regardless of the tools you used.**

When using AI to support your writing, you are asked to follow the “Read and Revise Until You Own (RRUYO)” rule because they will be held responsible for the content as explained here: <https://www.digitvibe.com/essence-of-writing/>

Your writing will be assessed as part of your grade in the following assessments:

- Participation (posting and commenting on the online forum)
- Lab assignments
- Final project

**Activities Outside of Class Time**

Although the final project may require group work outside of class time, regularly scheduled classes have precedence over any outside of class activities.

**Email Communication**

Please specify the course code in the subject line of all emails (ENTI 333 / 633). The instructor may not respond to emails that do not contain this information.

Although email is commonly used by students to communicate with their instructors, in some cases it may limit the effectiveness of communication and may not be the best way for me to answer student questions. If the student or instructor feels that communicating via email is not optimal, they may request a telephone call or personal meeting.

**Internet & Electronic Communication Devices**

It is recommended to bring a laptop to class in order to work with the digital tools discussed in class. During this class students will have the opportunity to utilize their personal electronic devices as learning tools to facilitate certain class activities. If your cell phone, tablet, or laptop does not become a distraction, it is fine for you to use it in class. The instructor may request that you put the device away if it is being used for activities unrelated to the class

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|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Academic Integrity and Rigor</b> | Academic integrity and rigor are critical components of a University degree. Academic integrity is the foundation of the development and acquisition of knowledge and is based on values of honesty, trust, responsibility, and respect. The Haskayne School of Business values ethical leadership and personal integrity, and expects its faculty, staff, and students to live these values. In the online environment, certain additional measures will be put in place to help safeguard the integrity of online assessments and the intellectual property of the instructors. |
| <b>Course Outline Part B</b>        | <p>The Course Outline Part B contains more generalized information for Haskayne and the University. You are responsible for reading and understanding all content in both parts of the outline.</p> <p>Part B can be found <a href="#">Here</a>.</p>                                                                                                                                                                                                                                                                                                                              |

## Class Schedule & Topics

Please note that lecture topics and readings are tentative and subject to change. The dates of assessments will not be changed.

Important dates (e.g. Block Week, Lecture start dates, Reading Week, etc.) can be found at the following web site: <https://calendar.ucalgary.ca/acadsched>

| COURSE SCHEDULE                   |                                                                                                                                                                                                 |                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| DATE                              | DETAILS                                                                                                                                                                                         | Notes                                                                                         |
| <b>Monday<br/>May 11, 2026</b>    | Understanding the Technology<br>Model and Tool Landscape<br>Lab 1 (AI Coding)                                                                                                                   | <b>Always bring your<br/>laptop to class<br/>Lab 1 Due by 11:59pm</b>                         |
| <b>Tuesday<br/>May 12, 2026</b>   | Prompt Engineering and Machine Psychology<br>Ethics of Generative AI and Superintelligence<br>Retrieval Augmented Generation (RAG) and Fine-Tuning<br>Autonomous Agents and Multi-Agent Systems |                                                                                               |
| <b>Wednesday<br/>May 13, 2026</b> | Building with Generative AI: No-Code and AI Coding<br>Getting Started on the Group Project<br>Lab 2 (Requirements Engineering)<br>Team Formation<br>Work on Group Project                       | <b>Lab 2 Due by 11:59pm</b>                                                                   |
| <b>Thursday<br/>May 14, 2026</b>  | Possible Guest Speaker or Panel<br>Work on Group Project                                                                                                                                        | <b>You may need time to<br/>work on the group<br/>project even outside of<br/>class hours</b> |
| <b>Friday<br/>May 15, 2026</b>    | First Half of Day:<br>Work on Group Project and Prepare Presentation<br>Second Half of Day:<br><b>Group Project Final Presentations</b>                                                         |                                                                                               |
| <b>Sunday<br/>May 17, 2026</b>    | <b>Group Project Blog Post and Github Links Due<br/>Last Chance to Contribute to Discovery Forum</b>                                                                                            |                                                                                               |



## Appendix: Details of Assignments

### The Discovery Forum:

A discussion board on D2L is dedicated to discoveries made by students in the class. Activity on this discussion board accounts for **most of your participation grade** in the class. This activity should include all three of the following:

1. **Posting:** post about things you discovered in the world of generative AI beyond what you learn in class. This can be a new app or tool, a learning resource (book, article, website, discord group, twitter thread, etc.), a prompting technique or template you found online, or something new you discovered yourself about the behavior of generative AI or an interesting way to use it. Pretty much anything about generative AI that students in the class would find useful is eligible as a post. Your posts must include more than just a link and also include your own thoughts on why the material is useful. A minimum of 10 posts is expected from each student over the course of the term.
2. **Voting:** upvote the best posts by others in the forum so that discovering the best material is made easier for others. A minimum of 20 votes is expected from each student over the course of the term.
3. **Commenting:** provide constructive commentary or feedback on posts made by others. A minimum of 10 comments is expected from each student over the course of the term.

You can contribute to the forum throughout the term up until the last day of class. Note that careless or unprofessional activity on the forum will be penalized, as will activity that is disrespectful or abusive.

### Group Project: AI-Assisted Software Development

The final project is a major deliverable worth 40% of your final grade where student teams (as assigned by the program) will develop a software application using modern AI-assisted development tools. This project combines practical development experience with business thinking, allowing you to create working software without needing prior programming experience. The best projects will be those that effectively leverage AI's strengths while maintaining a manageable scope that can be completed within the tight timeframe.

#### Team Formation Requirements

- Teams should be composed of 5-6 members
- Teams cannot be composed of all graduate students or all undergraduate students
- Teams cannot be all exchange students
- Exceptions may be granted on a case by case basis at the discretion of the instructor

#### Project Components and Grading

- Quality of Github & Documentation (5%): Professional structure and documentation provided on GitHub for the software application.
- Final Presentation (20%): This will be used to judge the quality of your work in the final project
- Blog Post (15%): A publicly published blog post or LinkedIn article documenting your journey and showcasing your work. Must include a video walkthrough of your app and the quality of the video counts as part of this grade component.

#### Core Requirements

Your team must build an application that:

- Solves a clear business problem or addresses a market opportunity (you can use AI to ideate, best to have a real client or target customer in mind, and please make sure the market for this kind of app is not already saturated or that at least you have a unique twist on the idea that is differentiated.)
- The app must be based on solid market research, ideation and requirements engineering such that the evidence you bring for why you thought this app is worth building is convincing. You can leverage AI to

help as much as you like, but real connections and conversations with real people and organizations with real pain points is the ultimate source for customer discovery and validation. Beware that AI tools sometimes make an idea seem more attractive than it really is by not comparing it to the right alternatives, not being aware of recent developments in that industry, or just not being critical enough.

- Is developed primarily using AI-assisted development tools (such as Replit.com, Antigravity.google, Cursor.com, Lovable.dev, v0.dev, Rork.app and Bolt.New). Use trial versions if you need to, but I recommend a paid plan on Replit.com (a month subscription for at least one member of the group should be enough, and if budget is a concern or you want to run concurrent experiments, try Antigravity). Using regular AI tools (ChatGPT, Claude, Gemini) to produce code and then copying and pasting it into a compiler is NOT sufficient for this exercise. The point is to learn to work with a specialized AI coding environment.
- Is properly documented on GitHub with AI assistance (including README, license file, and clear and comprehensive documentation of what the app is, what features it has, and its technical components)

### Technology Recommendations

- A web or mobile application built with html / css / javascript and the React framework is highly recommended as it works well with current AI tools and has excellent support, but other frameworks are acceptable.
- Deployment (making the working app available on the cloud) is optional but encouraged.
- Making your Github repository public with an open source license is required but exceptions may be allowed with proper justification if you want to keep the code private.

### Development Approach

You will use a combination of AI tools to develop your application. The course will cover various tools and approaches, but at minimum your development process should include:

1. Using AI tools like ChatGPT or Claude for initial ideation, requirements engineering, prototyping and component development
2. Using a proper AI coding tool for the main development
3. Leveraging AI for code documentation and GitHub repository setup

### Blog Post Requirements

Knowing that your name will be publicly associated with the project is used as a pedagogical tool here because it elevates the stakes and motivates high quality work. The blog post should be 1000-3000 words, published publicly on LinkedIn or a similar platform (not private or password protected), and must include:

- Acknowledgment of the course (Master of Management program at Haskayne School of Business, University of Calgary)
- Clear listing of all team members
- Problem statement and business case
- Overview of the solution and key features, including the tools used in the development process
- Discussion of the development process and how AI tools were used, with examples and screenshots of some of the prompts used.
- Lots of screenshots from the app user interface (do NOT include screenshots of the code)
- An embedded video tutorial showing a narrated screen recording of using the app (ideally no more than 3 minutes).
- Lessons learned and reflections on AI-assisted development
- CAUTION: a pdf file or a canva file or an infographic is NOT considered a blog post. A substack post, medium post, or linkedin article (but not linkedin post) is considered a blog post.

### Important Notes

- Teams are encouraged to validate their project idea with the instructor

- The project should be original work specific to this course and not building on work previously done outside of this course by any team member.
- The final presentation should effectively demonstrate the working application and your learning journey, and it will be used as a key point of judgement about the quality of your work. Cover everything that will help me judge the quality of the work, including a demo of the software in action (could be live or pre-recorded). I need to understand what you built, why you built it, how you built it, and how it works.
- The most important aspect of your presentation is to convey the value of the work that you did, not to spice up the presentation itself.
- It is encouraged but not required for all group members to participate in the presentation.

### **Submission Requirements**

By the deadline indicated in the course timetable, post to D2L:

1. A link to your GitHub repository
2. A link to your published blog post/article
3. A direct link to the video walkthrough
4. A link or attachment of your presentation slides

This is all a single deliverable to be handed in on a single deadline.